# STORY-009 Data Freshness Monitoring: evidence

STORY-009 works on the demo deployment, with synthetic data; it is not operating in production. Every profile save, eligibility check, application, case status change and assignment is timed from the start of the write until a separate read sees the new row version, the way the next request would. Staff get a Data Freshness page with p50, p95 and p99 per data type against a 30-second target for 95% of updates, an alert when a type falls behind, the slowest updates split into write time and wait time, and a Check now button that times a write end to end from the browser.

Screens 1 to 3 come from an automated run against a local copy of the deployed code, seeded through the API, because the demo site has no test case-manager accounts. Screen 3 follows a real slowdown: the run held one applicant's database row locked for 31 seconds while that applicant saved a profile. Screens 4 and 5 are the deployed demo site.

## Checks

- Demo deployment (scripts/verify-deploy.mjs): the API and the site serve 36a98bd; all 8 page paths serve the current build; database ok; 14 migrations applied; all 22 protected GET routes answer 401 without a token.
- On the demo deployment, signed in as a synthetic citizen (scripts/verify-live-citizen.mjs): all 13 staff GET routes answer 403, no staff links show, and the freshness page refuses the citizen.
- UI run (scripts/evidence/story-009.mjs): 8 checks passed, including the alert after the 31-second slowdown.
- API end to end, local (scripts/e2e): 105 checks passed: access 33, case workflow 26, compliance 30, freshness 16.
- Unit tests: 150 passed, including 12 new freshness tests and 117 route-access checks across all 40 routes.

## Screens

1. On target
2. Check now, end to end
3. A real slowdown raises the alert
4. Demo site: a citizen's dashboard
5. Demo site: a citizen opening /freshness

## Notes

- Measured on the demo deployment, which reads straight from its database with no cache or read replica yet, so a change is readable as soon as it is written. The same check, reading the row version on a separate request, will measure cache or replica lag when those arrive with AWS.
- CloudWatch metrics, named in the story's acceptance criteria, need the AWS environment (STORY-004, waiting on approval). Until then the same figures come from the demo's own measurements. Cache miss rate does not apply yet because the demo has no cache.
- Removed in this story: the /api/data and /api/latency placeholder endpoints. One reported data as fresh without measuring anything; another, a staff-only cleanup endpoint, put a request value straight into SQL. The route table now holds 40 routes.
- After a deploy, the site's CDN can serve the previous page for a deep link for up to 5 minutes. verify-deploy now waits until all page paths serve the current build before reporting a deploy as verified.
- All data shown is synthetic.

## 1. On target

Sam Ortiz, a case manager, opens Data Freshness after ordinary activity: 19 updates across four data types, all readable within 30 seconds, p95 172 ms. Each type shows when it last changed and how long its changes take.

**Data Freshness**
How long until each change is readable by the next request

Dashboard | Profile & Exemptions | Case Management | Compliance | Data Freshness | Logout

Demo environment. Synthetic data only. Do not enter a real SSN.

Last 24 hours | Last 7 days | Last 30 days | Check now

Within target: 100% of 19 updates were readable within 30 s (target 95%). p95 172 ms.

| Updates measured | p50 | p95 | p99 | Readable within 30 s |
|---|---|---|---|---|
| 19 | 8 ms | 172 ms | 172 ms | 100% |
| | | Target 30 s | | Target 95% |

### By data type

Each change is timed from the start of the write until a separate read sees it.

| Data type | Last update | Updates | p50 | p95 | p99 | Within 30 s | Status |
|---|---|---|---|---|---|---|---|
| Profile updates | 4 s ago took 2 ms | 3 | 3 ms | 172 ms | 172 ms | 100% | Within target |
| Eligibility checks | 4 s ago took 4 ms | 4 | 7 ms | 34 ms | 34 ms | 100% | Within target |
| Applications | 4 s ago took 6 ms | 4 | 7 ms | 14 ms | 14 ms | 100% | Within target |
| Case status and assignment | 4 s ago took 5 ms | 8 | 9 ms | 21 ms | 21 ms | 100% | Within target |
| End-to-end check | No updates yet | 0 | — | — | — | — | No updates yet |

### Slowest updates

Where the time went: the write itself, or the wait until the change was readable.

| When | Data type | Change | Write | Until readable | Total |
|---|---|---|---|---|---|
| 10/12/2026, 9:25:17 AM | Profile updates | Profile saved | 171 ms | 1 ms | 172 ms |
| 10/12/2026, 9:25:17 AM | Eligibility checks | Re-check after profile change | 32 ms | 2 ms | 34 ms |
| 10/12/2026, 9:25:18 AM | Case status and assignment | Status change | 20 ms | 1 ms | 21 ms |
| 10/12/2026, 9:25:17 AM | Case status and assignment | Status change | 15 ms | 1 ms | 16 ms |
| 10/12/2026, 9:25:17 AM | Applications | New application | 13 ms | 1 ms | 14 ms |
| 10/12/2026, 9:25:17 AM | Case status and assignment | Status change | 12 ms | 1 ms | 13 ms |
| 10/12/2026, 9:25:17 AM | Case status and assignment | Assignment | 9 ms | 2 ms | 10 ms |
| 10/12/2026, 9:25:18 AM | Case status and assignment | Assignment | 8 ms | 1 ms | 9 ms |
| 10/12/2026, 9:25:18 AM | Applications | New application | 9 ms | 1 ms | 9 ms |
| 10/12/2026, 9:25:18 AM | Case status and assignment | Status change | 7 ms | 1 ms | 8 ms |

Measured on the demo deployment. It reads straight from its database, with no cache or read replica yet, so a change is readable as soon as it is written; the same check will measure cache and replica lag once they exist. CloudWatch metrics come with the AWS environment (STORY-004).

## 2. Check now, end to end

Check now writes a record and times how long until a separate read sees it: readable 27 ms after the write began on the server. The browser's round trip was 50 ms, so about 23 ms was network and request handling.

# Data Freshness

How long until each change is readable by the next request

Dashboard | Profile & Exemptions | Case Management | Compliance | Data Freshness | Logout

Demo environment. Synthetic data only. Do not enter a real SSN.

Last 24 hours | Last 7 days | Last 30 days

Check now

**End-to-end check**
A new record was readable 27 ms after the write started on the server (26 ms to write, 1 ms until a separate read saw it). Round trip from this browser: 50 ms, so about 23 ms was network and request handling.

Within target: 100% of 19 updates were readable within 30 s (target 95%). p95 172 ms.

| Updates measured | p50 | p95 | p99 | Readable within 30 s |
|---|---|---|---|---|
| 19 | 8 ms | 172 ms | 172 ms | 100% |
| | | Target 30 s | | Target 95% |

### By data type

Each change is timed from the start of the write until a separate read sees it.

| Data type | Last update | Updates | p50 | p95 | p99 | Within 30 s | Status |
|---|---|---|---|---|---|---|---|
| **Profile updates** | 4 s ago took 2 ms | 3 | 3 ms | 172 ms | 172 ms | 100% | Within target |
| **Eligibility checks** | 4 s ago took 4 ms | 4 | 7 ms | 34 ms | 34 ms | 100% | Within target |
| **Applications** | 4 s ago took 6 ms | 4 | 7 ms | 14 ms | 14 ms | 100% | Within target |
| **Case status and assignment** | 4 s ago took 5 ms | 8 | 9 ms | 21 ms | 21 ms | 100% | Within target |
| **End-to-end check** | 0 s ago took 27 ms | 1 | 27 ms | 27 ms | 27 ms | 100% | Within target |

### Slowest updates

Where the time went: the write itself, or the wait until the change was readable.

| When | Data type | Change | Write | Until readable | Total |
|---|---|---|---|---|---|
| 10/12/2026, 9:25:17 AM | Profile updates | Profile saved | 171 ms | 1 ms | 172 ms |
| 10/12/2026, 9:25:17 AM | Eligibility checks | Re-check after profile change | 32 ms | 2 ms | 34 ms |
| 10/12/2026, 9:25:24 AM | End-to-end check | Write, then read back | 26 ms | 1 ms | 27 ms |
| 10/12/2026, 9:25:18 AM | Case status and assignment | Status change | 20 ms | 1 ms | 21 ms |
| 10/12/2026, 9:25:17 AM | Case status and assignment | Status change | 15 ms | 1 ms | 16 ms |
| 10/12/2026, 9:25:17 AM | Applications | New application | 13 ms | 1 ms | 14 ms |
| 10/12/2026, 9:25:17 AM | Case status and assignment | Status change | 12 ms | 1 ms | 13 ms |
| 10/12/2026, 9:25:17 AM | Case status and assignment | Assignment | 9 ms | 2 ms | 10 ms |
| 10/12/2026, 9:25:18 AM | Case status and assignment | Assignment | 8 ms | 1 ms | 9 ms |
| 10/12/2026, 9:25:18 AM | Applications | New application | 9 ms | 1 ms | 9 ms |

Measured on the demo deployment. It reads straight from its database, with no cache or read replica yet, so a change is readable as soon as it is written; the same check will measure cache and replica lag once they exist. CloudWatch metrics come with the AWS environment (STORY-004).

## 3. A real slowdown raises the alert

One applicant's database row was held locked for 31 seconds while that applicant saved a profile. Profile updates' p95 rose to 31.1 s, over the 30 s target, and the page raised the alert; the slowest-updates list shows the time went into the write, not the wait to read. Every other data type stayed in milliseconds.

**Data Freshness**
How long until each change is readable by the next request

Dashboard | Profile & Exemptions | Case Management | Compliance | Data Freshness | Logout

Demo environment. Synthetic data only. Do not enter a real SSN.

Last 24 hours | Last 7 days | Last 30 days | Check now

Over the freshness target
Profile updates: p95 is 31.1 s, over the 30 s target.

| Updates measured | p50 | p95 | p99 | Readable within 30 s |
|---|---|---|---|---|
| 21 | 8 ms | 172 ms (Target 30 s) | 31.1 s | 95.24% (Target 95%) |

### By data type

Each change is timed from the start of the write until a separate read sees it.

| Data type | Last update | Updates | p50 | p95 | p99 | Within 30 s | Status |
|---|---|---|---|---|---|---|---|
| Profile updates | 2 s ago, took 31.1 s | 4 | 3 ms | 31.1 s | 31.1 s | 75% | Over target |
| Eligibility checks | 2 s ago, took 7 ms | 5 | 7 ms | 34 ms | 34 ms | 100% | Within target |
| Applications | 38 s ago, took 6 ms | 4 | 7 ms | 14 ms | 14 ms | 100% | Within target |
| Case status and assignment | 38 s ago, took 5 ms | 8 | 9 ms | 21 ms | 21 ms | 100% | Within target |
| End-to-end check | 33 s ago, took 27 ms | 1 | 27 ms | 27 ms | 27 ms | 100% | Within target |

### Slowest updates

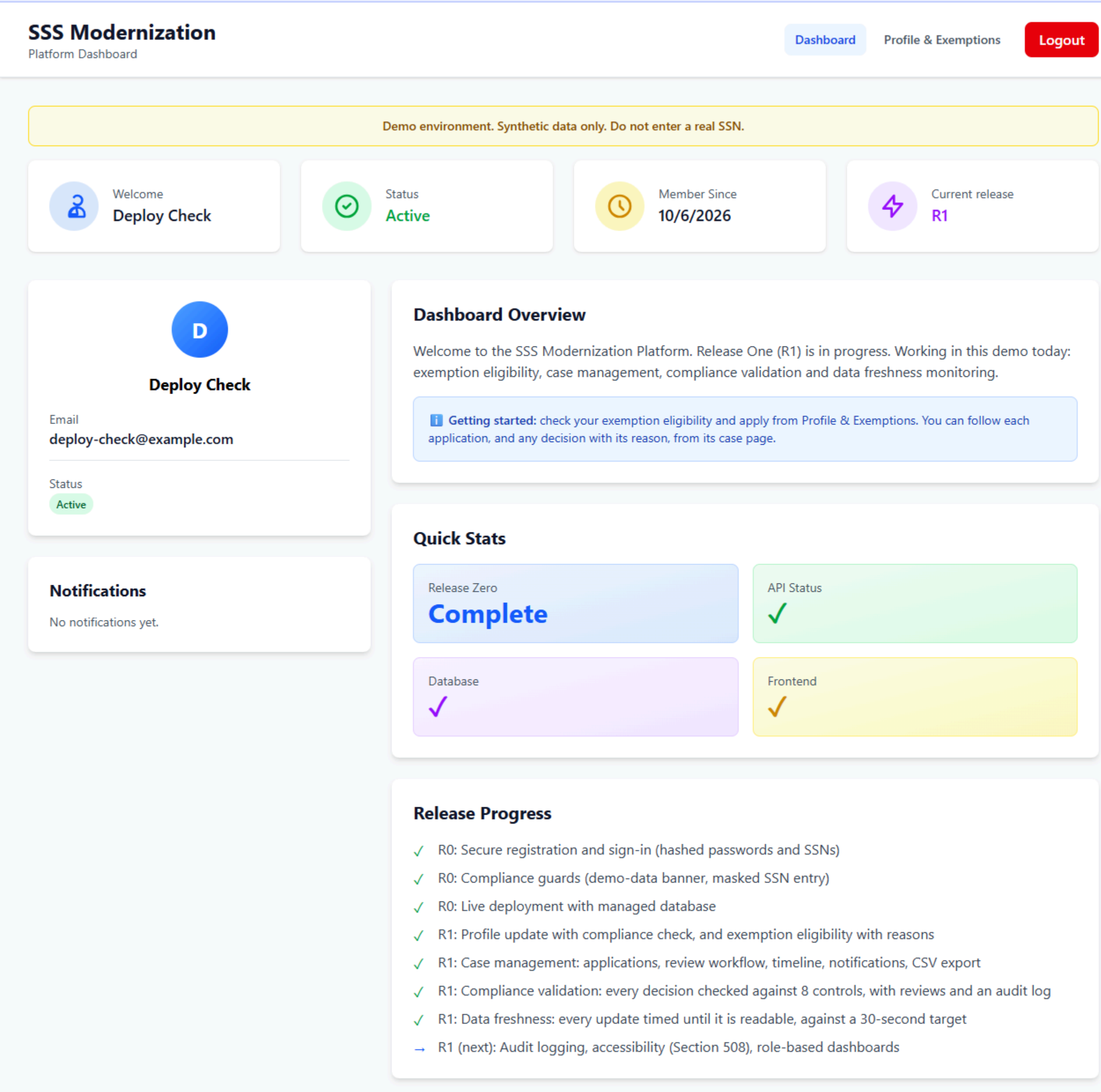
Where the time went: the write itself, or the wait until the change was readable.

| When | Data type | Change | Write | Until readable | Total |
|---|---|---|---|---|---|
| 10/12/2026, 9:25:55 AM | Profile updates | Profile saved | 31.1 s | 1 ms | 31.1 s |
| 10/12/2026, 9:25:17 AM | Profile updates | Profile saved | 171 ms | 1 ms | 172 ms |
| 10/12/2026, 9:25:17 AM | Eligibility checks | Re-check after profile change | 32 ms | 2 ms | 34 ms |
| 10/12/2026, 9:25:24 AM | End-to-end check | Write, then read back | 26 ms | 1 ms | 27 ms |
| 10/12/2026, 9:25:18 AM | Case status and assignment | Status change | 20 ms | 1 ms | 21 ms |
| 10/12/2026, 9:25:17 AM | Case status and assignment | Status change | 15 ms | 1 ms | 16 ms |
| 10/12/2026, 9:25:17 AM | Applications | New application | 13 ms | 1 ms | 14 ms |
| 10/12/2026, 9:25:17 AM | Case status and assignment | Status change | 12 ms | 1 ms | 13 ms |
| 10/12/2026, 9:25:17 AM | Case status and assignment | Assignment | 9 ms | 2 ms | 10 ms |
| 10/12/2026, 9:25:18 AM | Case status and assignment | Assignment | 8 ms | 1 ms | 9 ms |

Measured on the demo deployment. It reads straight from its database, with no cache or read replica yet, so a change is readable as soon as it is written; the same check will measure cache and replica lag once they exist. CloudWatch metrics come with the AWS environment (STORY-004).

## 4. Demo site: a citizen's dashboard

The deployed demo site lists data freshness as done in the release progress, and a citizen sees no staff links.



## 5. Demo site: a citizen opening /freshness

The page refuses, and the demo site's API answers 403 to a citizen on all 13 staff routes.

